

Four in, nineteen properties of general relativity out

what goes in

a Lorentzian signature

one shared metric

the value of G

the value of Λ

first two: what a metric theory is

last two: measured constants

**CPT of the universe,
fused with algebraic QFT**

the metric is never quantised

what comes out

- the field equations
- a horizon temperature
- horizons at every boost
- an area law for entropy
- the same coefficient everywhere
- the transverse involution
- G does not run
- $G > 0$, gravity attracts
- Λ as an integration constant
- null-focusing surfaces
- four dimensions, no others
- the classical-quantum divide
- black-hole thermodynamics
- a hot bang
- the weak equivalence principle
- the contracted Bianchi identity
- matter conservation
- universality of local horizons
- the two sheets' opposite time